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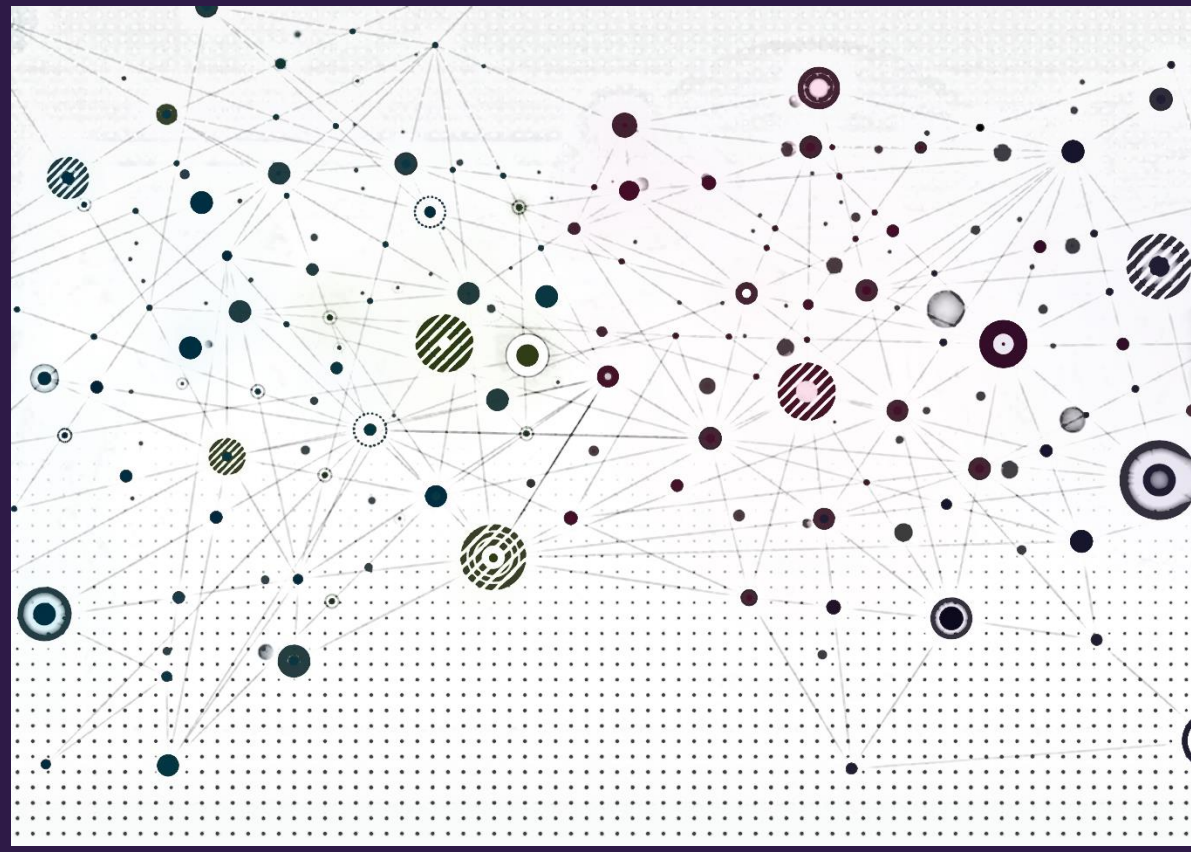
Classical Graph Representations and Properties



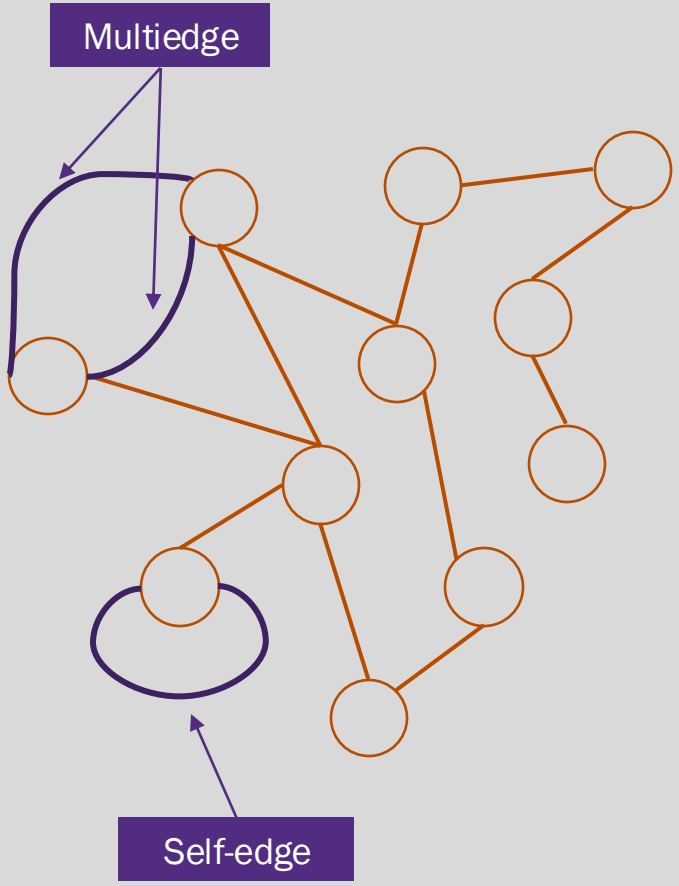
Overview

- Adjacency matrix
- Node degree and network density (sparsity)
- Graph Laplacian
- Walks and paths
- Centrality measures
- Node groups
- Node similarity metrics

Adjacency matrix

What is a graph?

- A graph, $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, is defined by a set of nodes, $\mathcal{V}$, and a set of edges, $\mathcal{E}$. An edge from node $u \in \mathcal{V}$ to node $v \in \mathcal{V}$ is denoted as $(u, v) \in \mathcal{E}$.
- n denotes the number of nodes
- m denotes the number of edges
- *multiedge* occurs when there are multiple edges between the same node
- *Self-edge* (or self-loop) occurs when a node is connected to itself

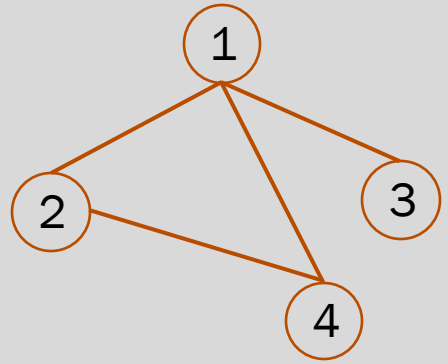


Adjacency matrix for a simple undirected network

- Consider a "simple" undirected network with unweighted edges and no self-loops, and no multi-edges
- Label the nodes with integers $1 \cdots n$
- The ***adjacency matrix,*** $\mathbf{A}$, is the $n \times n$ matrix with elements a_{ij}

$$a_{ij} = \begin{cases} 1 \; if \; there \; is \; an \; edge \; between \; nodes \; i \; and \; j \\ 0 \; otherwise \end{cases}$$

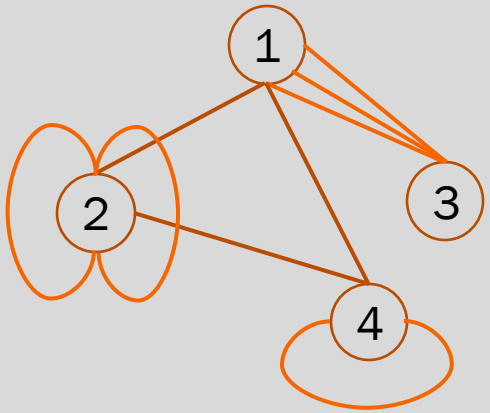
  - Diagonal elements are all zero (no self-loops)
  - Matrix is symmetric, $a_{ij} = a_{ji}$



	1	2	3	4
1	0	1	1	1
2	1	0	0	1
3	1	0	0	0
4	1	1	0	0

Adjacency matrix with self-loops and multiedges

- Multiedge is represented by setting a_{ij} and a_{ji} equal to the multiplicity of the edge
  - Example: a triple edge between nodes i and j is represented by $a_{ij} = a_{ji} = 3$
- Self-edge (node connected to itself)
  - *Single self-edge is represented by setting $a_{ii} = 2$. Why not 1?
  - Multiple self-edges are represented by setting $a_{ii} = 2 \times$ multiplicity (i.e., twice the number of self-edges on node i)

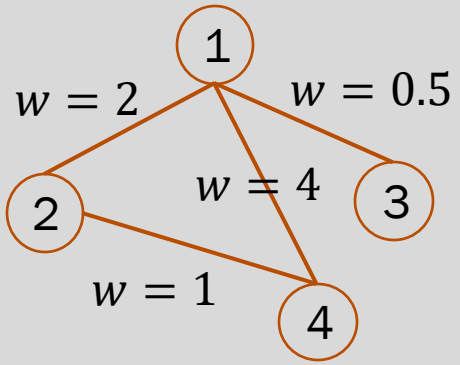


	1	2	3	4
1	0	1	3	1
2	1	4	0	1
3	3	0	0	0
4	1	1	0	2

*In directed networks, self-edges are indicated by $a_{ii} = 1$ (see slide 9)

Adjacency matrix on weighted networks

- Weights may be assigned to edges to indicate the "strength" of the connection
  - Data flow capacity between networked devices
  - Energy flow in a physical system
- Weights are typically positive but not required (e.g., social networks with adversarial relationships)
- Weights are represented in the adjacency matrix by setting $a_{ij} = a_{ji} = w$ where w is the weight of the edge between i and j
- Adjacency matrix is still symmetric



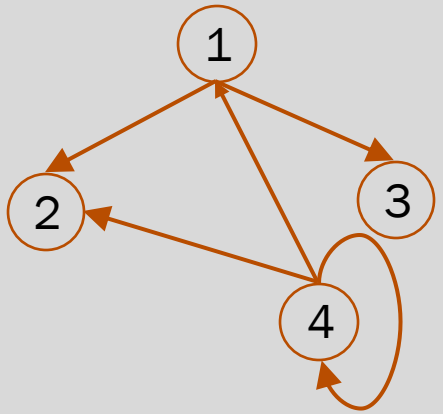
	1	2	3	4
1	0	2	0.5	4
2	2	0	0	1
3	0.5	0	0	0
4	4	1	0	0

Adjacency matrix on directed networks

- Edges in a directed network are one-way connections
  - A directed edge from node i to j does not imply an edge from j to i
- The ***adjacency matrix,*** **A**, of a **directed network** is the $n \times n$ matrix with elements a_{ij}

$$a_{ij} = \begin{cases} 1 \; if \; there \; is \; an \; edge \; \boldsymbol{from \; j \; to \; i} \\ 0 \; otherwise \end{cases}$$

- Adjacency matrix of a directed network is generally asymmetric, $a_{ij} \neq a_{ji}$
- Self-edges (unweighted) are represented by a 1 (not 2 as in undirected network)
- Rules for multi-edges and weighted edges are the same as undirected networks



	1	2	3	4
1	0	0	0	1
2	1	0	0	1
3	1	0	0	0
4	0	0	0	1

Node degree & network density (sparsity)

Node degree for undirected networks

- The degree, k_i, of node i in an **undirected** network is the number of edges incident to i
  - NOT the number of nodes connected to it
  - A multiedge adds its multiplicity to the node degree

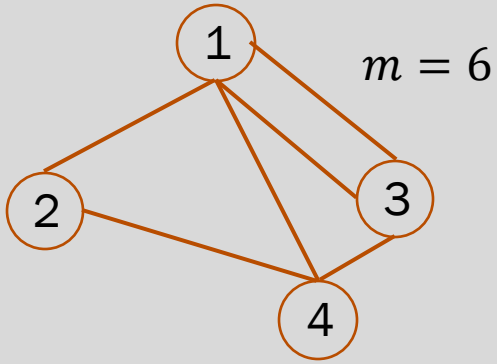
$$k_i = \sum_{j=1}^{n} a_{ij}$$

- An undirected network with m edges has $2m$ edge ends (i.e., points incident on a node) and hence

$$2m = \sum_{j=1}^{n} k_i = \sum_{ij} a_{ij}$$

- The mean degree over all nodes in the network is

$$c = \frac{2m}{n}$$



Node	Degree
1	4
2	2
3	3
4	3

$$\sum_{j=1}^{4} k_i = 12 = 2m$$

$$c = \frac{12}{4} = 3$$

Node degree for directed networks

- Two degree types for each node in a directed network
  - *in-degree* is the number of edges entering the node
  - *out-degree* is the number of edges leaving the node
- Recalling that $a_{ij} = 1$ in the adjacency matrix of a directed network if there is an edge from j to i, then

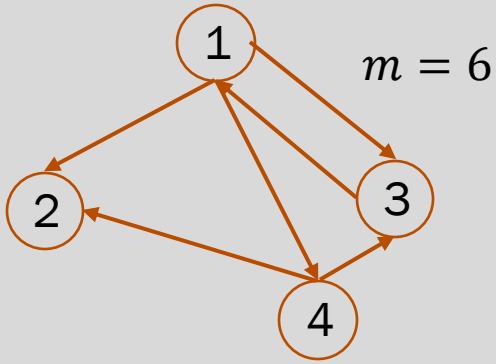
$$k_i^{in} = \sum_{j=1}^{n} a_{ij} \text{ and } k_i^{out} = \sum_{j=1}^{n} a_{ji}$$

- The total number of edges, m, is equivalently given by any of

$$m = \sum_{i=1}^{n} k_i^{in} = \sum_{j=1}^{n} k_j^{out} = \sum_{ij} a_{ij}$$

- The mean *in-degree* is <u>always</u> equal to the mean *out-degree*

$$c_{in} = c_{out} = c = \frac{m}{n}$$



Node	*In-degree*	*out-degree*
1	1	3
2	2	0
3	2	1
4	1	2

$$\sum_{i=1}^{4} k_i^{in} = \sum_{j=1}^{4} k_j^{out} = 6$$

$$c = \frac{6}{4} = 1.5$$

Network density

- The ***maximum*** number of edges in a simple, undirected network with n edges is $m_{max} = \binom{n}{2} = \frac{n(n-1)}{2}$ (Why?)
- The *density*, ρ, of the network is the fraction of m_{max} edges that are present. That is for a network with m edges:

$$\rho = \frac{m}{m_{max}} = \frac{2m}{n(n-1)} = \frac{c}{n-1} \approx \frac{c}{n}$$

- Probability that a pair of randomly selected nodes is connected
- *Dense network* – if the network density, ρ, remains non-zero as the number of nodes, n increases, it is considered *dense*
- *Sparse network* – if the network density, ρ, approaches zero as the number of nodes, n increases, it is considered *sparse*

Network density and growth

- In a dense network where ρ is constant as $n \to \infty$, the average degree, c, grows linearly
  - Recall for large n, that $\rho = \frac{c}{n}$
  - To remain constant, we require $c = \rho n \longrightarrow m = \frac{\rho n^2}{2}$
- In a sparse network, the average degree, c, can grow
  - sublinearly (sparse) OR
  - remains constant in which case $\rho \to \frac{1}{n}$ (extremely sparse)

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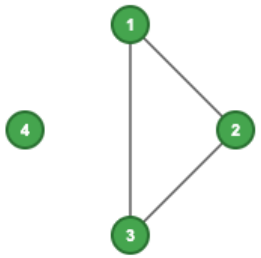
Edge growth in a dense network $\rho = {}^{1}/_{2}$

Graph with 4 nodes

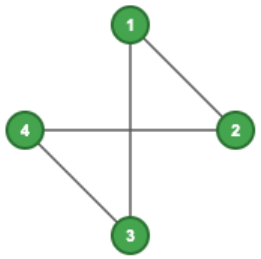
Exact Formula
Edges: 3
Density: 0.500

Asymptotic Formula
Edges: 4
Density: 0.667
Error: 33.3%

Exact



Asymptotic

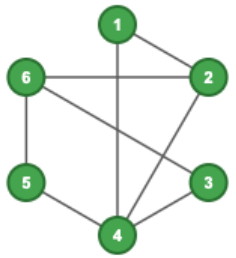


Graph with 6 nodes

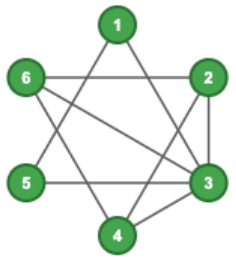
Exact Formula
Edges: 8
Density: 0.533

Asymptotic Formula
Edges: 9
Density: 0.600
Error: 12.5%

Exact



Asymptotic

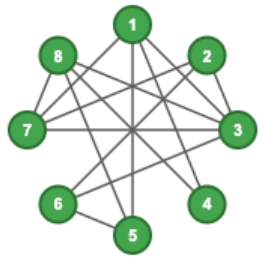


Graph with 8 nodes

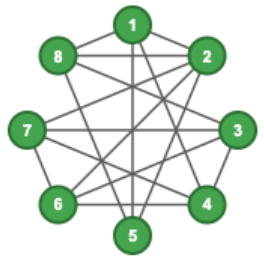
Exact Formula
Edges: 14
Density: 0.500

Asymptotic Formula
Edges: 16
Density: 0.571
Error: 14.3%

Exact



Asymptotic

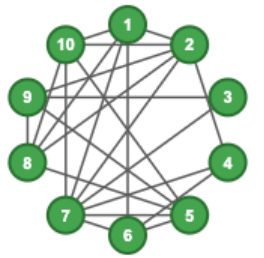


Graph with 10 nodes

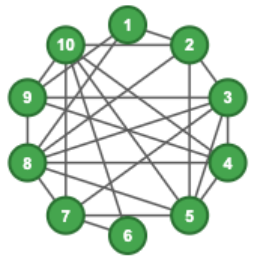
Exact Formula
Edges: 23
Density: 0.511

Asymptotic Formula
Edges: 25
Density: 0.556
Error: 8.7%

Exact



Asymptotic



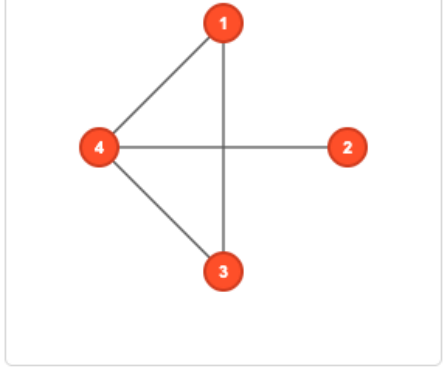
https://ml4g.masinolab.com/topics/network-metrics/constant_edge_density.html

Edge growth in a sparse network $c = 2$

Sparse Graph: 4 nodes

Network Properties:
Nodes (n): 4
Edges (m): 4
Max possible edges: 6
Edge density (p): 0.500
Average degree: 2.0

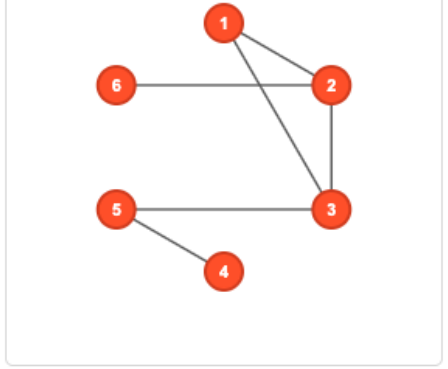
p = c/n = 2/4 = 0.500



Sparse Graph: 6 nodes

Network Properties:
Nodes (n): 6
Edges (m): 6
Max possible edges: 15
Edge density (p): 0.333
Average degree: 2.0

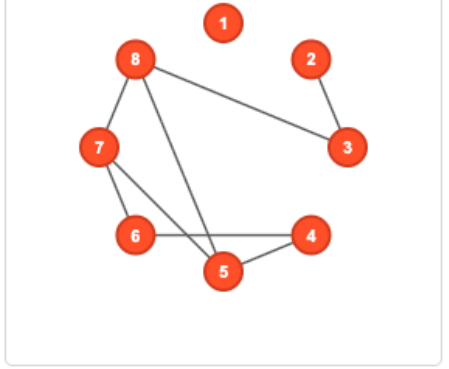
p = c/n = 2/6 = 0.333



Sparse Graph: 8 nodes

Network Properties:
Nodes (n): 8
Edges (m): 8
Max possible edges: 28
Edge density (p): 0.250
Average degree: 2.0

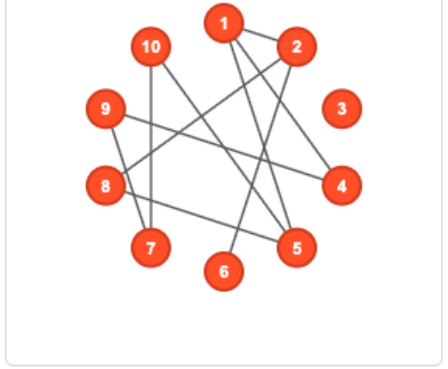
p = c/n = 2/8 = 0.250



Sparse Graph: 10 nodes

Network Properties:
Nodes (n): 10
Edges (m): 10
Max possible edges: 45
Edge density (p): 0.200
Average degree: 2.0

p = c/n = 2/10 = 0.200



https://ml4g.masinolab.com/topics/network-metrics/sparse_network_constant_degree.html

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Graph Laplacian

Graph Laplacian

- The graph Laplacian, $\mathbf{L}$, is an $n \times n$ matrix with elements

$$l_{ij} = k_i \delta_{ij} - a_{ij} = \begin{cases} k_i \; if \; i = j \\ -1 \; if \; i \neq j \; and \; there \; is \; an \; edge \; from \; i \; to \; j \\ 0 \; otherwise \end{cases}$$

where k_i is the degree of node i and δ_{ij} is the Kronecker delta (1 if $i = j$ and 0 otherwise

- Matrix form $\mathbf{L} = \mathbf{D} - \mathbf{A}$ where

$$\mathbf{D} = \begin{pmatrix} k_1 & 0 & \cdots \\ 0 & k_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

- For weighted networks, substitute the weighted adjacency matrix and $k_i = \sum_j a_{ij}$
- No extensions for networks with self-edges or directed networks

Graph Laplacian properties

- Every row (and column) of the graph Laplacian, $\mathbf{L}$, sums to zero

$$\sum_j l_{ij} = \sum_j \left(k_i \delta_{ij} - a_{ij}\right) = k_i - k_i = 0$$

- All eigenvalues are ≥ 0
- There is always at least one zero eigenvalue - for a connected graph there is exactly one, otherwise there is one zero eigenvalue for each component
- The *algebraic connectivity* or *spectral gap* is the value of the smallest non-zero eigenvalue

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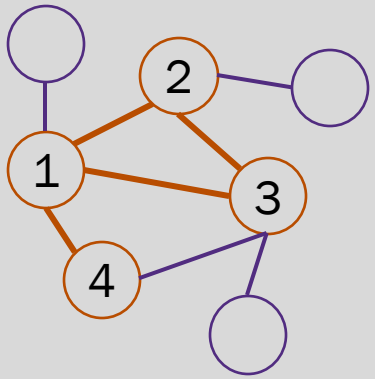
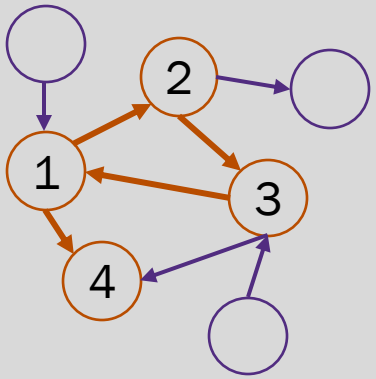
Walks and paths

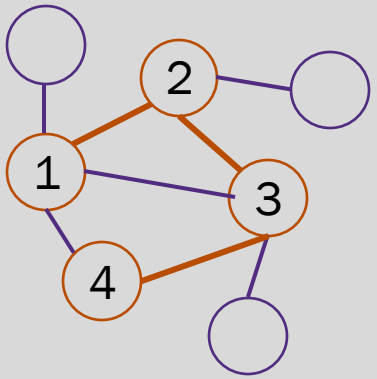
Walks and paths

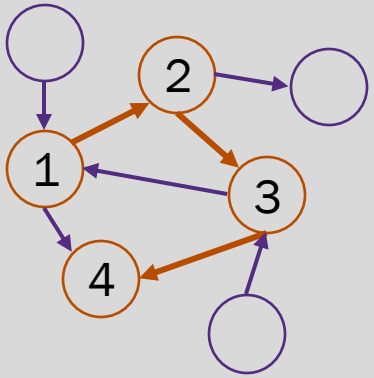
- A *walk* is any sequence of nodes such that every consecutive pair is connected by an edge
  - In a directed network, each edge in a walk must follow the edge direction
- A *path* is a *walk* that does not intersect itself (i.e., all nodes in a path are distinct)





The *walk* $\{(1,2),(2,3),(3,1),(1,4)\}$ traverses the network from node 1 to 4. It intersects itself at node 1 and is, therefore, **not a path**.





The *walk* $\{(1,2),(2,3),(3,4)\}$ traverses the network from node 1 to 4. It does **not** intersect itself. **This walk is also a path.**

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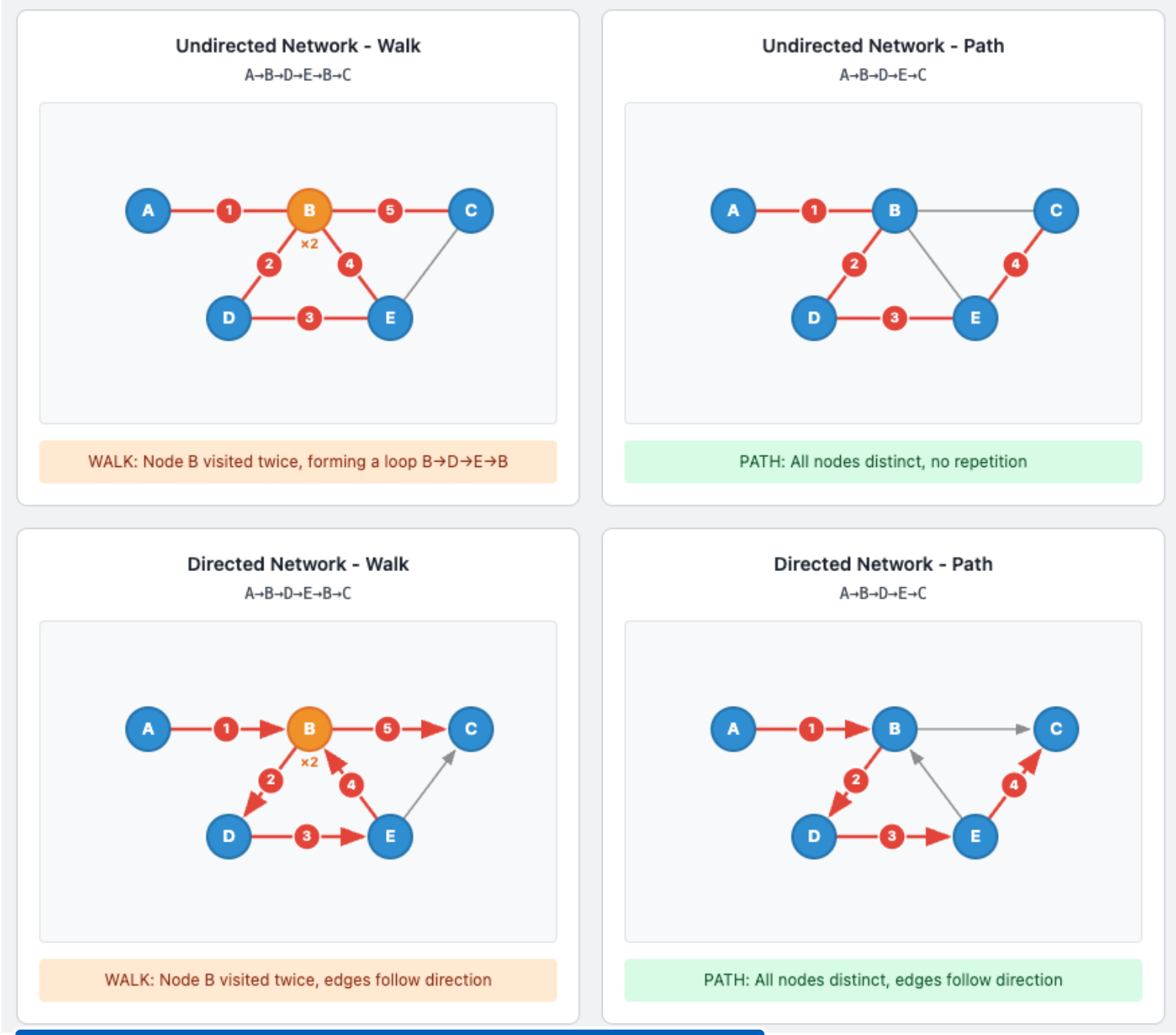
Network walks & paths

- Walk – sequence of nodes where consecutive pairs are connected by edges
- Path – a walk where all nodes are distinct



Legend
- Regular node
- Repeated node
- Highlighted sequence

https://ml4g.masinolab.com/topics/network-metrics/walk-path-visualization.html

Survey – participation is optional

Please take a moment to complete this survey

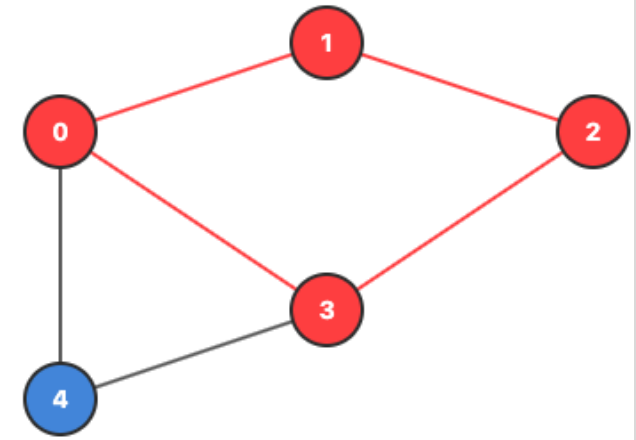
Walk length and walk counts

- The walk length is the number of edges (not nodes) in the walk
- The total number of **walks** of length r between nodes i and j is

$$w_{ij}^{(r)} = [\boldsymbol{A}^r]_{ij}$$

where $[\boldsymbol{A}^r]_{ij}$ is the element a_{ij} in the adjacency matrix to the r^{th} power

- Note, walks have direction even in undirected networks. Walk 1⟶2 is distinct from 2⟶1



3	0	2	1	1
0	2	0	2	1
2	0	2	0	1
1	2	0	3	1
1	1	1	1	2

Walks of length two from node 0 to 2 are highlighted in red. The corresponding value of $[A^2]_{0,2}$ is equal to 2 indicating these are the only two paths.

https://ml4g.masinolab.com/topics/network-metrics/network-path-visualization.html

Shortest paths

- A *shortest path* is the shortest walk (i.e., traverses the fewest edges) between a pair of nodes
  - Such a walk is necessarily a path (Why?)
- The *network diameter* is the length of the longest shortest path (i.e., among all shortest paths between every pair of nodes, the diameter is the longest of these)
- Knowledge of shortest path distributions can inform graph neural network design (e.g., how many layers) (future lecture)

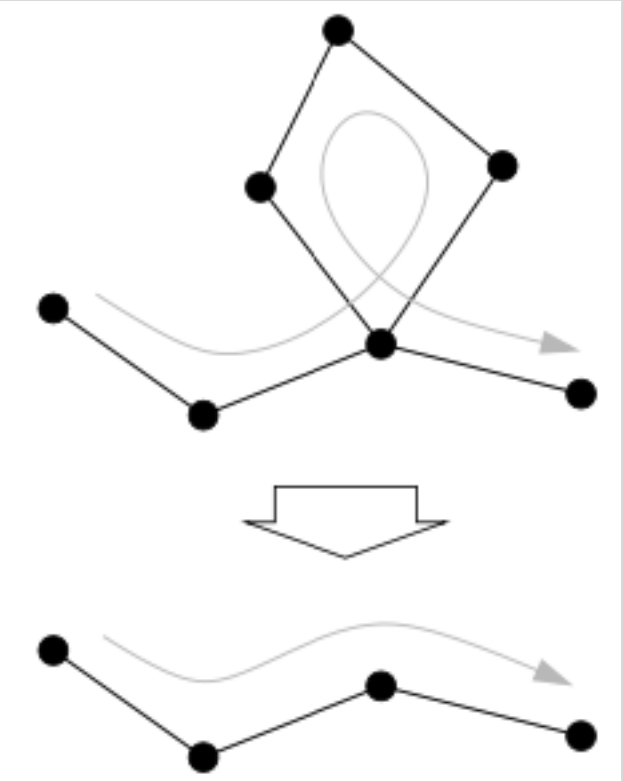


Image credit: Newman 2018

Independent paths and connectivity

- *Edge independent paths* – two paths connecting nodes i and j are **edge** independent if they share no edges
- *Node independent paths* - two paths connecting nodes i and j are **node** independent if they share no node
- *Edge connectivity* is the number of edge independent paths between a pair of nodes.
- *Node connectivity* is the number of node independent paths between a pair of nodes.

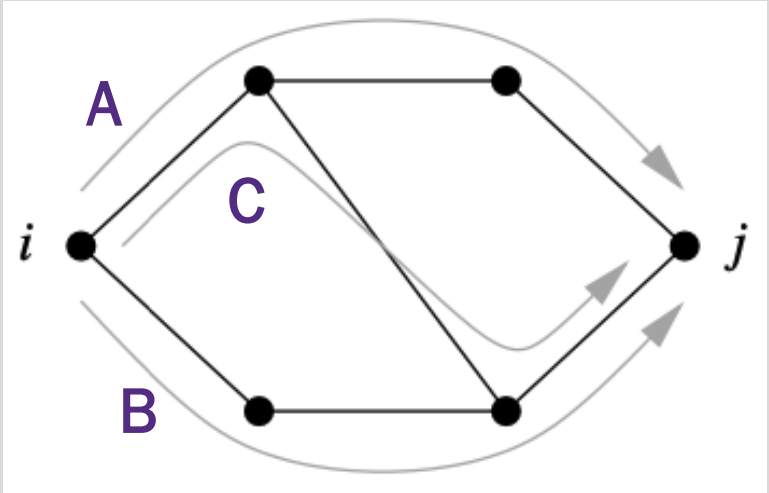


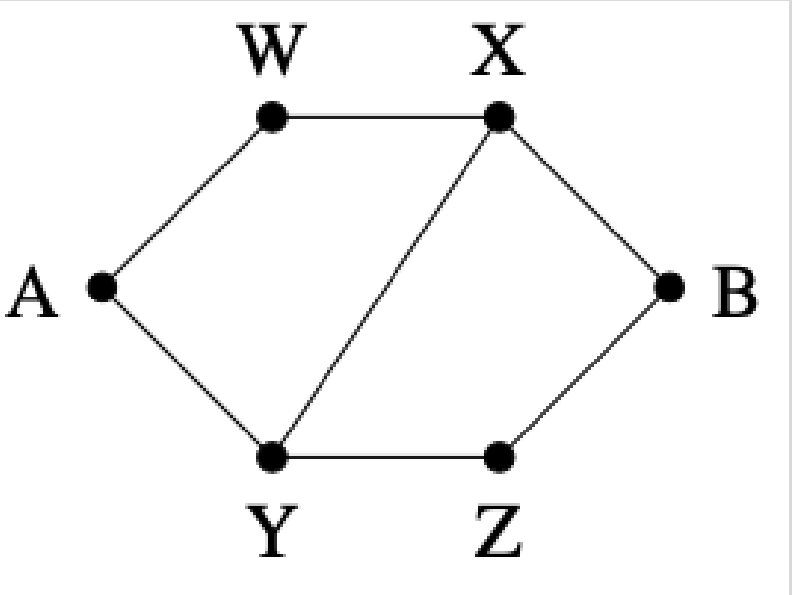
Image credit: Newman 2018

- Paths **A** and **B** are node and edge independent from each other
- Path **C** is NOT node or edge independent from paths **A** or **B**
- Node and edge connectivity between i and j is 2

Bottlenecks and cut sets

- A *node cut set* is a set of nodes whose removal will disconnect a specified pair of nodes
  - A cut set with a single node is a bottleneck
  - We'll see later that bottlenecks can be a problem in GNNs
- An *edge cut set* is a set of edges whose removal will disconnect a specified pair of nodes
- *Minimum cut set* is the smallest cut set that will disconnect a pair of nodes
  - *Menger's theorem:* the minimum cut set is equal to the number of independent paths between the nodes

Image credit: Newman 2018

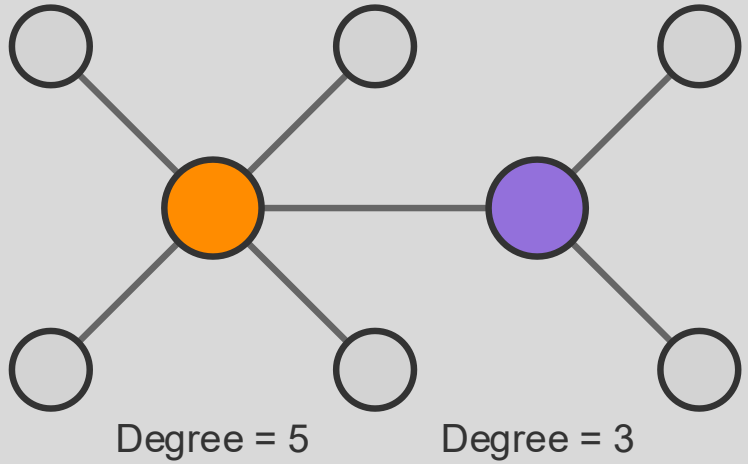


- Two independent paths from **A** to **B**: {A, W, X, B}, and {A, Y, Z, B}
- Minimum cut sets are {W, Y} and {X, Z}, {X, Y}
- Why isn't {W, Z}?

Centrality measures

Centrality concept

- Centrality metrics are intended to measure the importance of a node in a network
- Node degree can be considered a centrality measure where nodes with higher degree are considered more important
  - Doesn't consider importance of neighbors
  - Doesn't consider role in network structure
- There are many others to discuss



Eigenvector centrality for undirected networks

- In many networks, a node is important if it is connected to other important nodes
- Consider an undirected network. Let's define the centrality, x_i, of node i to be *proportional* to the centralities of its neighbors

$$x_i = \kappa^{-1} \sum_{j=1}^{n} a_{ij} x_j$$

where κ^{-1} is the inverse proportionality and a_{ij} is the i, j element of the adjacency matrix

- **So we need the centrality measures to compute the centrality measures? How do we proceed?**

Eigenvector centrality for undirected networks

- We can rewrite $x_i = \kappa^{-1} \sum_{j=1}^{n} a_{ij} x_j$ in matrix form $\mathbf{x} = \kappa^{-1}\mathbf{A}\mathbf{x}$ which is equivalent to **eigenvalue problem:**

$$\mathbf{A}\mathbf{x} = \kappa\mathbf{x}$$

- Select the leading eigenvector of $\mathbf{A}$ (the one corresponding to the largest positive eigenvalue)
- Set κ to the value value of the leading eigenvalue
- Centrality x_i of node i is the i^{th} element of the leading eigenvector

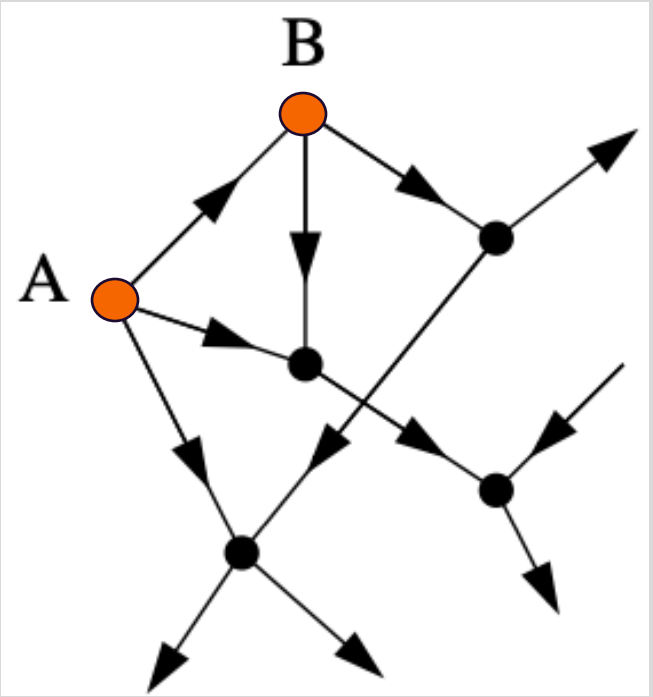
Linear Algebra for the win!!

Linear algebra is fundamental to machine learning. Graph learning is no exception. If you want a linear algebra refresher, I suggest Dr. Gilbert Strang's MIT's open course.

Eigenvector centrality on directed networks

- Adjacency matrix of a directed network generally has *left* and *right* eigenvectors
- The *right* eigenvector indicates the importance of node i based on the importance of nodes that point to it.
- Limitations
  - Only nodes in a strongly connected component of two or more nodes (or its out component) will have non-zero eigenvector centrality
  - A node could have a high in-degree but zero eigenvector centrality

Image credit: Newman 2018



- Node A has zero eigenvector centrality because it has no nodes pointing to it
- Node B is pointed to by node A only, so it also has zero eigenvector centrality

PageRank

- Addresses limitations of eigenvector centrality on directed networks by
  - Adding a small constant importance, β, to **every** node
  - Scaling the contribution of neighbor nodes importance by the inverse of their out-degree

$$x_i = \alpha \sum_j a_{ij} \frac{x_j}{k_j^{out}} + \beta$$

- In matrix form (setting $\beta = \mathbf{1}$)

$$\mathbf{x} = \alpha \mathbf{A}\mathbf{D}^{-1}\mathbf{x} + \mathbf{1} = (\mathbf{I} - \alpha \mathbf{A}\mathbf{D}^{-1})^{-1}\mathbf{1}$$

- The scaling term α must be chosen carefully.
  - If it's too small, every node's PageRank approaches β
  - If it's too large, $\mathbf{I} - \alpha \mathbf{A}\mathbf{D}^{-1}$ becomes singular
  - Undirected networks choose $\alpha < 1$, directed network requires experimentation

Google

- PageRank was developed by Google for web page ranking
- Still plays a central role in Google search, though other factors are used
- α=0.85 – no one knows how this was selected!

Betweenness Centrality

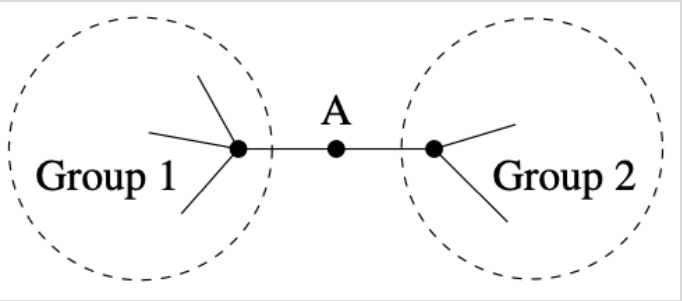
- Measures the extent to which a node is on shortest paths between other nodes
- The betweenness centrality, x_i, of node i is

$$x_i = \sum_{st} \frac{n_{st}^i}{g_{st}}$$

where the sum is over all shortest paths for all node pairs $\{s, t\}$, n_{st}^i is the number of shortest paths from s to t that contain i, and g_{st} is the number of shortest paths from s to t

- Removal of nodes with high betweenness centrality will most disrupt ”flow” (of information, water, energy, etc.)
- May be convenient to normalize by $1/n^2$
- Can be applied to directed and undirected networks

Image credit: Newman 2018



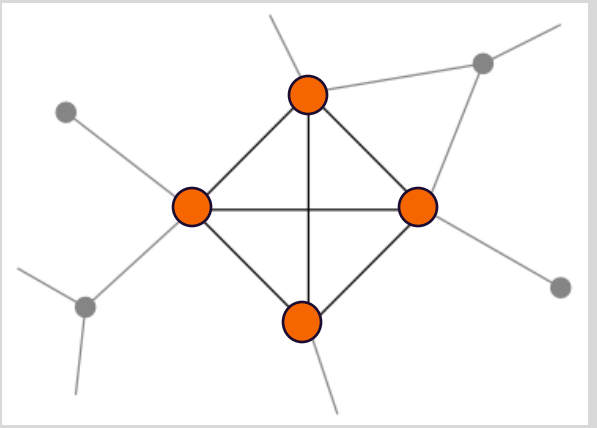
- Node A has low degree (only 2 neighbors)
- All paths between nodes in Group 1 and those in Group 2 must pass through A
- A will have high betweenness centrality

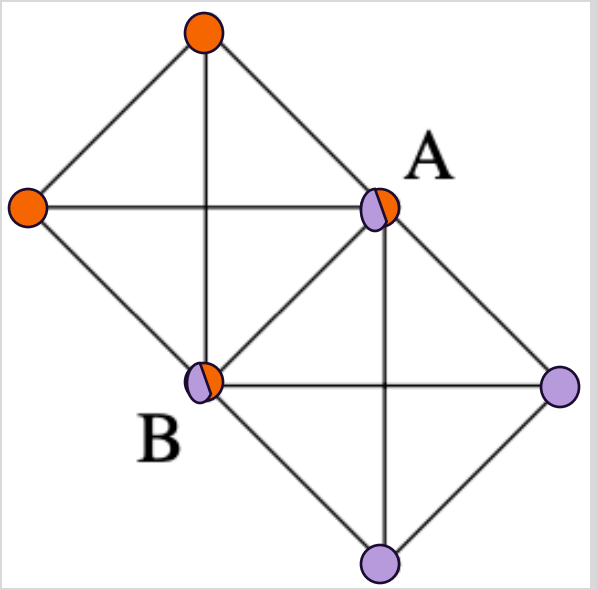
Node groups

Cliques

- A *clique* is a set of nodes in which every node is connected to every other node within an undirected network
- Cliques may overlap, i.e., some nodes in one clique may belong to another clique
- A clique found in a sparse network may indicate that the clique nodes are more similar than randomly selected nodes
  - Family members in a social group
  - Groups of interacting proteins in cellular networks
  - Financial entities that interact through transactions

Image credit: Newman 2018





- A clique of 4 nodes (top)
- Two cliques with overlapping nodes (bottom)

k-cores

- A *k-core* is a set of connected nodes (every node is reachable from every other node) where each node is linked to at least k of the other nodes
- Less stringent grouping than a clique
- *k-cores* are nested, i.e., the nodes in the *k-core* are in the nodes in the *(k-1)-core*

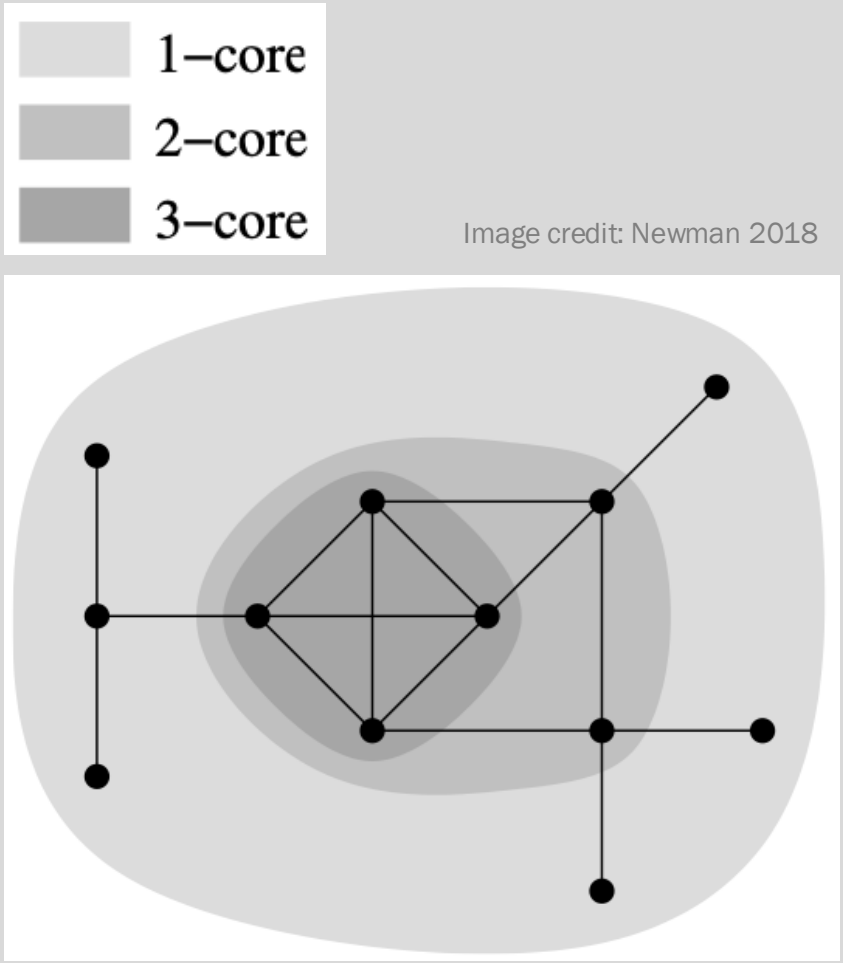


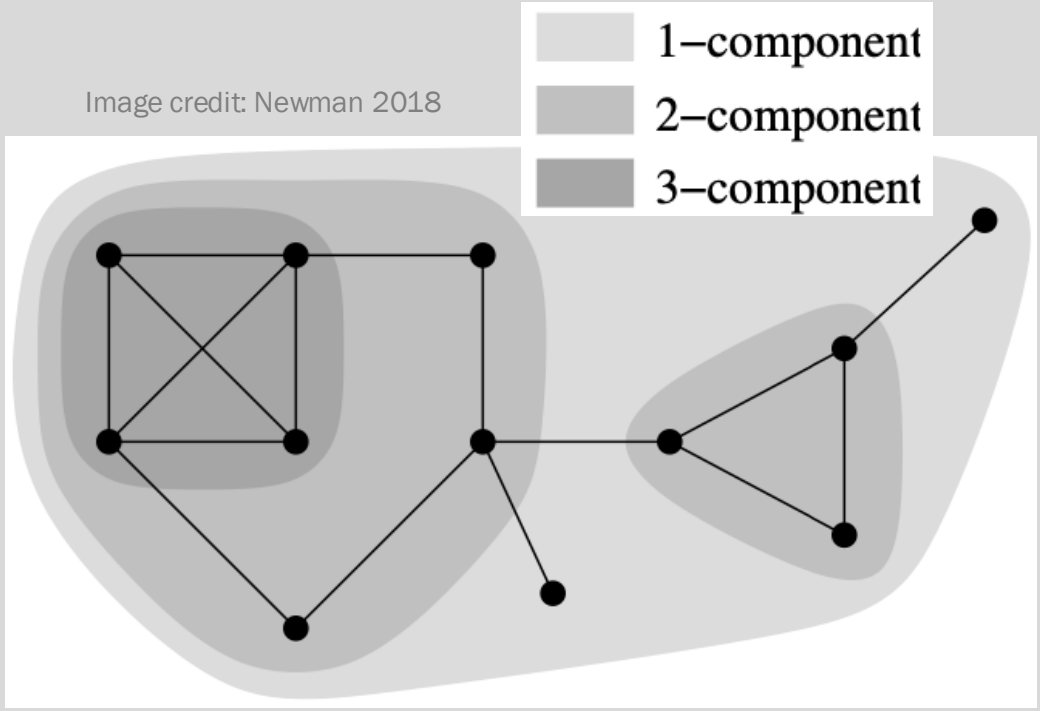
Image credit: Newman 2018

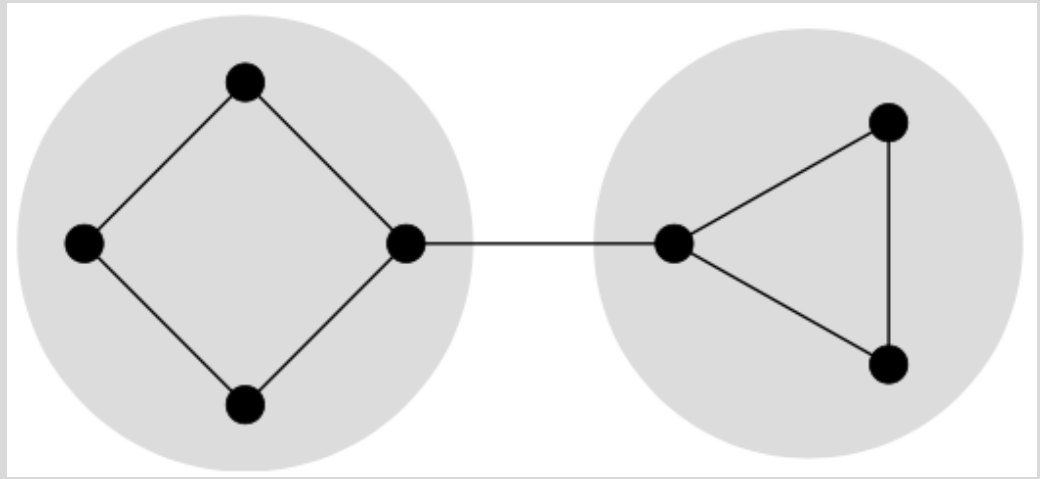
- All nodes in the network together form the 1-core
- The 2-core nodes are a subset of the 1-core nodes
- The 3-core nodes are a subset of the 2-core nodes
- No cores with $k > 3$

k-components

- A *k-component* is a set of nodes where every node is reachable from every other node by at least *k* **node-independent** paths
- No pair of nodes can be disconnected without removing less than *k* other nodes
- *k-components* are nested like *k-cores*
- Distinct from *k-cores*

Image credit: Newman 2018





- K-components are different than k-cores
- (Bottom) Two 2-components (shaded) and one 2-core (entire network)

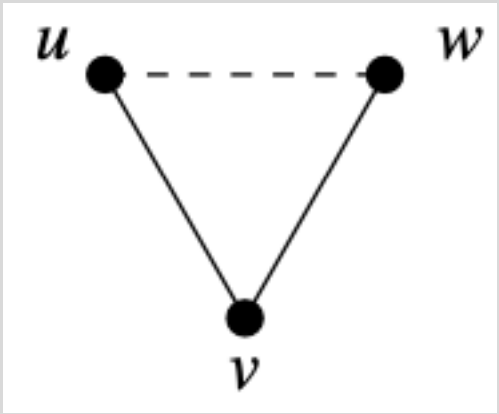
Clustering coefficient

- Consider an undirected network in which node u is connected to node v and node v is connected to node w
  - Yields the 2-edge path $u \rightarrow v \rightarrow w$
  - If u is also connected to w we say the path is closed
- The *clustering coefficient*, C, is the fraction of all such paths of length two that are closed

$$C = \frac{Number\ of\ closed\ paths\ of\ length\ two}{number\ of\ paths\ of\ length\ two}$$

$$C = \frac{(number\ of\ triangles) \times 6}{number\ of\ paths\ of\ length\ tow}$$

Image credit: Newman 2018



- Social network studies have found clustering coefficients around 0.1
- Random connections in the studied networks would yield clustering coefficients <0.001
- Suggests two people will be connected if they have a shared connection

https://ml4g.masinolab.com/topics/network-metrics/clustering-coefficient-demo.html

Homophily

- *Homophily* in networks is the tendency of nodes to be connected to similar nodes
- Also called assortative mixing
- Often based on node features, e.g., class membership
- *Degree homophily* (assortative mixing by degree, i.e., nodes are connected to nodes with similar degree) is derived from network structure and is defined by

$$\frac{1}{2m}\sum_{ij}\left(a_{ij}-\frac{k_i k_j}{2m}\right)k_i k_j$$

which is the covariance of node degree across all node pairs in the network

Node similarity metrics

Node similarity

- Here, we focus on similarity measures that only consider network structure.
- Later, we'll see methods that incorporate node features

Cosine similarity

- Consider two vectors $\mathbf{x}$ and $\mathbf{y}$. The cosine of the angle, θ, between them is

$$cos\theta = \frac{\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}||\mathbf{y}|}$$

- Consider an undirected, unweighted network. Let the i^{th} and j^{th} rows, $\mathbf{a}_i$ and $\mathbf{a}_j$, of the adjacency matric be vectors, their cosine similarity, σ_{ij}, is

$$\sigma_{ij} = \frac{\mathbf{a}_i \cdot \mathbf{a}_j^T}{|\mathbf{a}_i||\mathbf{a}_j|} = \frac{n_{ij}}{\sqrt{k_i k_j}}$$

  where n_{ij} is the number of shared nodes between nodes i and j
- Structural equivalence metric
- Value lies in $[0,1]$ (assuming non-negative vector values)
- Widely used in machine learning to compare embedding vector similarity

Other structural equivalence similarity metrics

- Jaccard coefficient

$$J_{ij} = \frac{n_{ij}}{k_i + k_j - n_{ij}}$$

- Pearson correlation coefficient

$$r_{ij} = \frac{(\mathbf{a}_i - \bar{\mathbf{a}}_i) \cdot (\mathbf{a}_j - \bar{\mathbf{a}}_j)}{\sqrt{(\mathbf{a}_i - \bar{\mathbf{a}}_i)^2} \sqrt{(\mathbf{a}_j - \bar{\mathbf{a}}_j)^2}}$$

Where $\bar{\mathbf{a}}_j$ is the mean of row j